

Parent Functions III: $1/x$ and Asymptotes - Practice

Algebra II Honors | Unit 1 | Lesson 6 Practice | TEK A2.2.A

Objective: I can prove $1/x$ is odd, describe its end behavior precisely, and find asymptotes of related rational rules.**Section 1 — Recall**

1. Complete: $f(x) = 1/x$ at $x = -4, -1, 1, 4$.

2. Give the domain of $f(x) = 1/x$ in set notation.

Section 2 — Apply

3. Write, in end-behavior notation, what happens as $x \rightarrow \infty$ for $f(x) = 1/x$, and as $x \rightarrow -\infty$.

 $x \rightarrow \infty$: $f(x) \rightarrow$ _____ $x \rightarrow -\infty$: $f(x) \rightarrow$ _____

4. Prove algebraically that $m(x) = 5/x$ is odd.

 $m(-x) =$ _____**Section 3 — Challenge**

5. Find all vertical asymptotes of $k(x) = 1/(x^2 - 5x + 6)$ by factoring the denominator, and give the domain in interval notation.

Factors: _____ VAs: _____

Domain: _____

6. A rational function $r(x) = 1/(x - a)$ always has domain "all reals except $x = a$." Explain why no value of the constant a could ever make this function's domain all real numbers, connecting your answer to the definition of a vertical asymptote.

Section 4 — Explain It

7. A classmate claims that because $1/(x^2 + 1)$ has no vertical asymptote (since $x^2 + 1$ is never 0), it must therefore have a horizontal asymptote different from $y = 0$, "since it needs an asymptote somewhere." Is the classmate correct? Explain what actually determines a horizontal asymptote.
